InSynBio

Clinical Data-Guided AI *Biotherapeutic Strategy System*

Fusing immunology logic, clinical data benchmarking, and private AI inference to deliver actionable drug design — not just isolated algorithm scores.

Presenter: Jing Huang, Ph.D.

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Field: AI-Powered Drug Design

FOUNDER & VISION

Jing Huang, Ph.D.

Bridging the gap between empirical immunology and structural AI algorithms to build the next generation of biotherapeutic engineering tools.

■ Academic Background

- **Columbia University**
- **Rockefeller University**
- **Albert Einstein College of Medicine**

■ What We Do

"We don't just supply generic AI algorithms.

By anchoring cutting-edge ML models (AlphaFold, Diffusion, LMs) to true clinical wet-lab databases, we deliver actionable, experimentally-validated decision support for complex macromolecules."

PLATFORM SCOPE

Mastering Next-Gen Modalities

Beyond standard antibodies, we leverage clinical-data-guided AI to design and evaluate the full spectrum of modern immunotherapy.

mAbs & VHH
AI-driven humanization and stability optimization to minimize IND-stage failure.

Bispecifics
Optimizing format selection and geometry-controlled chain pairing precision.

ADC Portfolio
DAR optimization and structure-aware linker-payload compatibility design.

CAR-T / CAR-M
Modular binder selection from 237-item library to overcome T-cell exhaustion.

mRNA & AI Vaccines
LNP-delivery optimization and rational antigen scaffolding for potent response.

AI Evaluation
End-to-end design & CMC flagging using Clinical Data Benchmarks.

"We provide Decision Architecting: leveraging the logic of AI to move beyond trial-and-error across all immune drug modalities."

CHALLENGES WE SOLVE EVERY DAY

Three Problems You Shouldn't Face Alone

These are the most common pain points our clients bring to us — and we have a clear answer for each.

"Humanization killed our affinity."

The most common setback we hear.

What we do: Structural deviation analysis + golden VH-VL pairing predicts framework risk before substitution — ensuring 100% CDR binding retention.

"We don't know if this molecule is developable."

Algorithm scores don't reveal where IND risk actually hides.

What we do: 15 parameters benchmarked against 1,142 marketed drugs — each flagged pass, caution, or fix with specific guidance.

"ADC/CAR design feels like guesswork."

Complex modalities shouldn't depend on luck and mass screening.

What we do: Target biology logic + clinical database matching delivers evidence-based design recommendations — not random exploration.

HOW WE WORK

You Receive a Report. Not a Software License.

Submit your sequence. In 3–5 days, you receive a structured analysis report ready for your team to act on — no tools to learn, no data scientists needed.

Every deliverable includes:

① Item-by-item risk checklist — pass / caution / fix, with mutation-level guidance.
② Candidate ranking with percentile vs. 1,142 approved drugs.
③ Actionable next steps: which experiments to prioritize, which variants to test first.

Buying AI Software

You still need: Bioinformatics engineers, domain experts to interpret outputs, and weeks to integrate. Result: Numbers on a screen — but no clear next step.

InSynBio Service

Submit sequence → receive report in 3–5 business days. Expert interpretation: Every report backed by immunology + AI specialists, directly supporting your go/no-go decision.

Pure Wet-Lab Screening

Time cost: Each iteration takes weeks, failures repeat without root-cause clarity. Result: Potentially 9+ months before discovering the molecule cannot be developed.

OUR REFERENCE BASELINE

Your Molecule Is Compared to Real Drugs

Not guessed by an algorithm — benchmarked against 1,000+ approved and clinical-stage drugs, property by property. That's the foundation of every assessment we deliver.

1,142 Clinical Antibody Sequences	**458** Engineered Antibodies (AbRef-458)	**138** Clinical ADA Immunogenicity Records	**100** Clinical ADC Project Data	**237** CAR-T Design Components
842 Clinical Germline Database	**134+** Bispecific CMC Benchmarks	**109** MHC I/II Epitope Records	**42** Clinical VHH Reference Molecules	**21** Clinical Fc Engineering Records

DECISION MATRIX ENGINE

AbEngineCore — The 8-Series Decision Matrix

STRUCTURAL PREDICTION	MOLECULAR DOCKING	GENERATIVE DESIGN	STABILITY & AFFINITY
AlphaFold2	**HADDOCK3**	**ProteinMPNN**	**ThermoMPNN**
ABodyBuilder2	AF2-Multimer	ESM-IF1 · RFdiffusion	MM/GBSA · PRODIGY

IMMUNOGENICITY	SEQUENCE ANNOTATION	BIOLOGICAL FITNESS	VACCINE DESIGN SUITE
IEDB API	**ANARCI**	**AntiFold**	**NeoantigenScanner**
35 HLA-DRB1 Alleles	Kabat / IMGT Numbering	ESM-2	MultiEpitopeAssembler

"A unified 8-series toolstack providing end-to-end decision intelligence for all biotherapeutic modalities."

DEVELOPABILITY ASSESSMENT

Know Your Risks Before the Lab Does

We assess 15 parameters across 4 directions and tell you exactly where your molecule stands — before you commit to cell-line development or IND-enabling studies.

1. Primary Sequence

SOLVES: BASIC SOLUBILITY & STABILITY

5 Parameters: pI point, GRAVY hydrophobicity, Instability index, Net charge, SAP aggregation score.

2. TAP Guidelines

SOLVES: INDUSTRY OUTLIER RISK

4 Parameters: CDR Length, PSH (Hydrophobic patches), PPC/PNC (Charge patches), SFvCSP (Asymmetry).

3. Structural Risk

SOLVES: FOLDING & INTERFACE FLAWS

3 Parameters: VH-VL Angle consistency, Contact Maps (Steric clashes), Vernier SASA exposure.

4. Chemical Snags

SOLVES: MANUFACTURING LIABILITIES

3 Parameters: Deamidation (NG/NS), Oxidation (M/W/H), Isomerization (DG/DS) sequence scans.

What You See in Your Report: The IDI Score

All 15 parameters roll up into a single percentile score against 1,142 approved drugs. You see exactly where your molecule ranks, which properties need attention, and — critically — what to do about them before spending another dollar in the lab.

15D

Clinically Grounded

IMMUNOGENICITY

ADA Immunogenicity — Multi-Dimensional Benchmarking

ADA is extremely difficult to accurately predict. A multi-dimensional comparison against analogous clinical molecules is the only reliable anchor — we refuse to mislead with 100% foresight.

■ Sequence Dimension

MHC-II 35 HLA-DRB1 T-Cell Epitope Scanning · Risk Site Curating Clustering · Parker Hydrophilicity Filtering

■ Structure + Germline

Historical ADA rates comparing analogous germlines · Surface Immunogenicity (SASA) · Germline Tolerance Validation

■ Clinical Verification

138 actual antibody drugs with clinical ADA data · multi-parameter benchmarking

View Report →

"We cannot guarantee absolute zero immunogenicity, but rigorously supported by the most compreh
ensive clinical data supply chain, we aim to minimize the relative risk to the industry's lowest achievable level."

Fig 6b — ROC · FH AUC 0.690

[Chart Figure]

Fig 6c — Confusion Matrix @0.646 · FH Sensitivity 100% Full Report →

[Chart Figure]

ENGINEERING PIPELINE

Standardized Engineering & SOP Matrix

1	→	2	→	3	→	4	→	5	→	6	→	7
Seq Analysis		Humanization		Structure & Epi.		Affinity VAM		CMC Check		Clinical Benchmark		Report Delivery

■ Intelligent Humanization

Machine Learning-driven selection + structural back-mutations ensure 100% CDR stability.

■ Virtual Affinity VAM

6-tool $\Delta\Delta G$ consensus validation for rapid, high-confidence binder optimization.

■ CDR Core Redesign

ProteinMPNN-driven novel topologies to circumvent IP moats and radicalize diversity.

Excellence Gallery — Real-world & PoC Portfolio

I. COMMERCIAL SUCCESS STORIES

Bispecific VHH Formats

pI engineering: 8.2 → 6.8 stability. 4.8× expression increment via GS-8 linker design.

View Comprehensive Report ■

Small Molecule VAM

Matching binder affinity for commercial assay thresholds; ΔΔG –5.53 kcal/mol yields.

View Comprehensive Report ■

PD-L1 Epitope Map

Unveiled sterical blocking domains; validated MOA with PRODIGY scores (–10.8 vs –7.3).

View Comprehensive Report ■

II. TECHNICAL VALIDATIONS (PoC)

Humanization SOP

muMAb4D5: 100% structural binding preservation using 842-profile germlines.

Validation Link ■

CMC 15D Benchmarking

Validated 15 core parameters; mapped to 1,142 clinical assets (IDI 77/100).

Validation Link ■

HER2 VHH Affinity VAM

Binding energy improved –3.32 kcal/mol via epistatic double mutant G49A+F112L.

Validation Link ■

CDR Core Redesign

ProteinMPNN IP circumvention; deriving novel backbone topologies.

Validation Link ■

PROVEN RESULTS II

Smart Engineering — Intelligent Drug Design Systems

III. INTELLIGENT DRUG DESIGN

ADC Decision Engine: CLDN18.2 Case

Navigating extreme Gastric Cancer heterogeneity. Optimizing for CLDN18.2 Moderate/Slow internalization & Low shedding based on target profiling metrics.

Case Study: Intelligent Design for Gastric Cancer ■

CAR-T Modality Architect

Precision modular signaling assembly using the InSynBio 237-binder library, optimizing for CAR durability against solid tumor microenvironments.

Case Study: Anti-CIDRα1 Dual-Binder CAR-Macrophage Design ■

Vaccine Design Suite

Rational antigen scaffolding & multi-allelic epitope assembly suites verified against 109 core MHC records and clinical immunogenicity evidence.

KRAS G12D Neoantigen Multi-Epitope mRNA Design — Case Study ■

IV. FUTURE FRONTIERS (ONGOING RESEARCH PROJECTS)

De Novo Protein Drug Design

- Synergistic generation via RFdiffusion, ProteinMPNN, and AlphaFold2.
- Exploring novel binding sequence spaces independent of existing scaffolds.
- Integrated workflow: De novo design + structural validation + druggability assessment.

[Status: Active R&D / Collaboration Exploration]

TCR-Restricted Epitope Immunopotentiation

- Binding energy optimization for TCR recognition of pMHC epitopes.
- Enhancement of T-cell immune response via multi-HLA coverage strategies.
- AI analysis and mutation design of the pMHC-TCR complex interface.

[Status: Active R&D / Collaboration Exploration]

HOW TO GET STARTED

Simple Process, 3 Steps to Launch

Per-Project Engagement

① Sign NDA — sequences are fully protected

② Submit sequences, receive structured report in 3–5 days

③ Debrief call to align on next experimental steps

+ 50% fee return for academic co-publications

Ongoing Advisory

→ Continuous pipeline support — analyze new candidates anytime

→ Join internal R&D meetings with real-time expert input

→ All sequences and data deleted after project closes

→ Scope adjusts flexibly to your pipeline stage

CONTACT US

Have a Molecule?
Send It Over.

Email us your sequence for a free preliminary assessment. No commitment, no pressure.

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